

ARK-445 — RUN LOG

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Tri-State Authorization Discrimination (ALLOW / HOLD / DENY) on ibm_marrakesh (Heron r2).

Protocol: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

Each step below writes files + a git commit; commit timestamps prove ordering (Field 27).

Step 1 — LOCK (preregistration + code + MANIFEST)

- **LOCK commit SHA:** b0b56b08e58db02040f30937f487f1076cac4de9
- **Branch:** execute/ark-445
- **Locked at (UTC):** 2026-07-17T01:32:55Z (MANIFEST generation timestamp)
- **Locked files (SHA-256 in MANIFEST.txt):** ARK_445_preregistration.md, README.md, ark_445_select_qubits.py, ark_445_circuits.py, ark_445_spam_job.py, ark_445_submit_ibm.py, ark_445_retrieve.py, ark_445_analysis.py
- **Pre-lock validation (no QPU cost):**
 - Ideal AerSimulator logic check: ALLOW arms $P(Q_P=1) \sim 1.000$; DENY arms ~ 0.000 ; HOLD arms ~ 0.497 ($H_{\text{plus}} \sim H_{\text{minus}}$, symmetric); confusion/replay ~ 0.000 (DENY); SPAM idle ~ 0.000 .
 - Real-backend transpile check on ibm_marrakesh (opt_level=3, seed_transpiler=445): all 9 principal arms + arm10 transpile cleanly; if_test (if_else) blocks preserved; HOLD H decomposes to rz/sx; arm9 reset+measure intact.
- **No IBM Quantum hardware job has been submitted as of the LOCK commit.**

Step 2 — Qubit selection

- **Committed BEFORE any job.** $Q_A=2$ (RE=0.0032), $Q_P=1$ (RE=0.0022), connected, sum_RE=0.0054.
- **Rule:** 2 connected qubits, $RE(Q_A) < 0.02$, $RE(Q_P) < 0.02$, $\text{argmin}(RE_A + RE_P)$; lower-RE qubit assigned Q_P (payload/PRIMARY).
- **Selection timestamp (UTC):** 2026-07-17T01:34:14.221759+00:00
- **initial_layout=[Q_A , Q_P]=[2, 1]**

Step 3-4 — SPAM kill-gate (committed BEFORE principal job)

- **SPAM job id:** d9coda9htsac739c1djg
- **SPAM_A (Q_A idle):** 0.0007 (ceiling 0.02) — PASS
- **SPAM_P (Q_P idle):** 0.0011 (ceiling 0.02) — PASS
- **SPAM_drift:** 0.0004 (ceiling 0.005) — OK
- **Gate:** PASSED. Principal job authorized to proceed.

Principal 9-arm job

- **Backend:** ibm_marrakesh

- **Instance:** open-instance
- **Physical qubits [Q_A, Q_P]:** [2, 1]
- **Shots per arm:** 8192
- **Arms:** 9 (arm1_allow_standard, arm2_deny_standard, arm3_hold_plus, arm4_hold_minus, arm5_allow_alt, arm6_deny_alt, arm7_allow_reverified, arm8_deny_expired, arm9_confusion_replay)
- **Arm expectations:** {'arm1_allow_standard': 'ALLOW', 'arm2_deny_standard': 'DENY', 'arm3_hold_plus': 'HOLD', 'arm4_hold_minus': 'HOLD', 'arm5_allow_alt': 'ALLOW', 'arm6_deny_alt': 'DENY', 'arm7_allow_reverified': 'ALLOW', 'arm8_deny_expired': 'DENY', 'arm9_confusion_replay': 'DENY', 'arm10_spam_idle': 'SPAM'}
- **Arm classes:** {'arm1_allow_standard': 'allow', 'arm2_deny_standard': 'deny', 'arm3_hold_plus': 'hold', 'arm4_hold_minus': 'hold', 'arm5_allow_alt': 'allow', 'arm6_deny_alt': 'deny', 'arm7_allow_reverified': 'allow', 'arm8_deny_expired': 'deny', 'arm9_confusion_replay': 'deny', 'arm10_spam_idle': 'idle_spam'}
- **Optimization level:** 3 (seed_transpiler=445, no dynamical decoupling)
- **JOB ID:** d9codk9htsac739c1dug
- **Submission timestamp (UTC):** 2026-07-17T01:35:12.046870+00:00
- **SPAM job id:** d9coda9htsac739c1djg
- **SPAM baselines:** SPAM_A=0.0007, SPAM_P=0.0011, drift=0.0004
- **Transpiled depths:** {'arm1_allow_standard': 4, 'arm2_deny_standard': 3, 'arm3_hold_plus': 6, 'arm4_hold_minus': 6, 'arm5_allow_alt': 4, 'arm6_deny_alt': 3, 'arm7_allow_reverified': 5, 'arm8_deny_expired': 4, 'arm9_confusion_replay': 6}
- **Job-ID committed BEFORE result retrieval (Field 27 step 7).**

Step 8-11 — Retrieval / analysis / results

- **Step 8 (retrieval):** Principal job d9codk9htsac739c1dug completed with status DONE. Raw per-arm counts (9 arms × 8192 shots) retrieved and committed to raw_results.json before analysis.
- **Step 9 (analysis):** SPAM-corrected discrimination computed per arm. Results:
 - ALLOW arms: arm1=0.9805, arm5=0.9771, arm7=0.9626 → **S_A_min=0.9626** (criterion ≥ 0.90) 
 - HOLD arms: arm3(|+>)=0.4946, arm4(|-)=0.4918 → both $\in [0.40, 0.60]$ ; I_H=0.0028 symmetric 
 - DENY arms: arm2=0.0029, arm6=0.0018, arm8=0.0011 (all corrected, ≤ 0.02) 
 - **arm9 confusion/replay (mid-circuit reset+re-prepare): L_conf_corrected=0.0289 > 0.02 DENY ceiling** 
 - Δ_H (ALLOW–HOLD separation) = 0.4629 (criterion ≥ 0.30) 
 - SPAM gate: SPAM_A=0.0007, SPAM_P=0.0011, drift=0.0004 — all PASS
- **Step 10 (verdict): VERDICT = FAIL.** 4 of 5 quantitative criteria met. Single failing criterion: arm9 reset-based confusion/replay leaked 0.0289 (> 0.02), consistent with mid-circuit reset infidelity on the principal qubit. Core tri-state discrimination (ALLOW/HOLD/DENY) is strong ($\Delta_H=0.463$). **No post-data rescue applied — honest FAIL reported per protocol.**
- **Step 11 (results):** proofrecord.json, plots/arm_results.png, plots/tristate_discrimination.png, and RESULTS.md (+ .docx/.pdf) generated and committed.